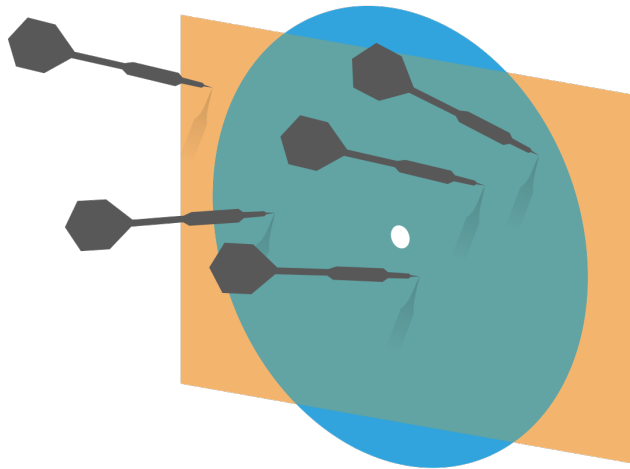


Probability and Statistics for Computer Science



“In statistics we apply probability
to draw conclusions from data.”
---Prof. J. Orloff

Credit: wikipedia

Last time

✱ Normal (Gaussian) distribution

Objectives

- * Sample mean

$$\frac{\sum_{i=1}^N x_i}{N}$$

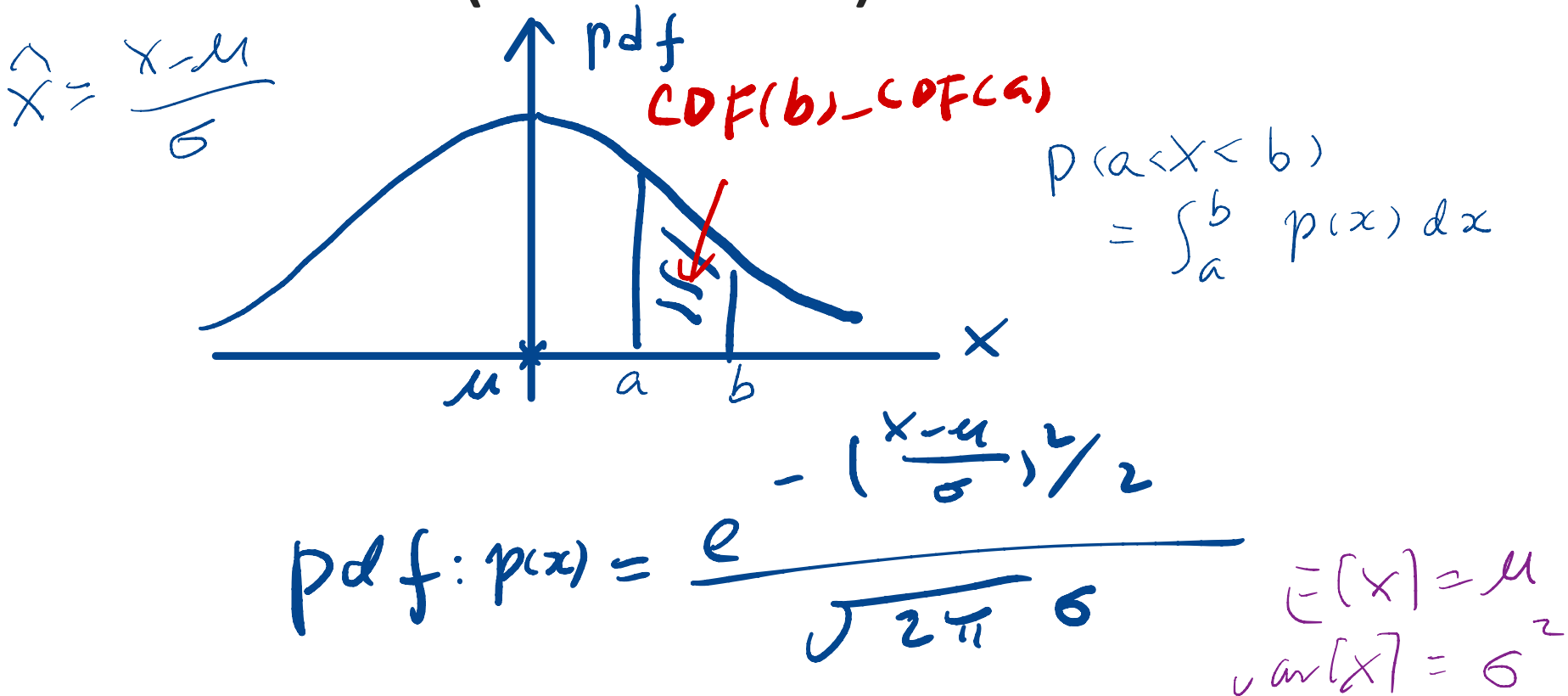
iid

- * confidence interval

- * t-distribution

Last time

✱ Normal (Gaussian) distribution



What's the $\Pr(|X| < 1.5)$?

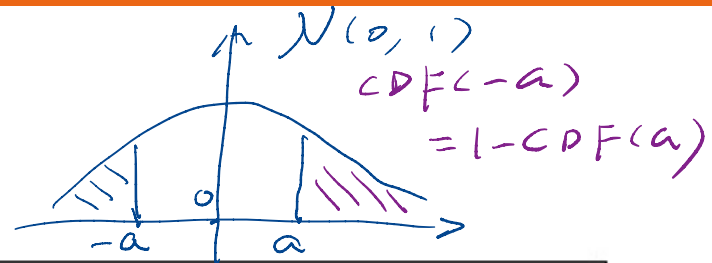
$$X \sim N(0, 1)$$

Q. Is the table with only positive x values enough?

A. Yes

B. No.

$P(|X| < 1.3) = 2(\text{CDF}(1.3) - 1) = 0.8064$



	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319

$X \sim N(0, 1)$

$$P(|X| < 1.3)$$

$$= P(-1.3 < X < 1.3)$$

$$= \text{CDF}(1.3) - \text{CDF}(-1.3)$$

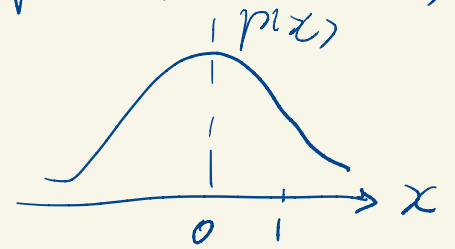
$$= \text{CDF}(1.3) - (1 - \text{CDF}(1.3))$$

$$= 2 \text{CDF}(1.3) - 1$$

$$= 2 \times 0.9032 - 1$$

$$= 0.8064$$

$$X \sim \mathcal{N}(0, 1)$$



$$p(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$$

$$\mu = 0, \sigma = 1$$

Motivation for drawing conclusion from samples

- ✱ In a study of new-born babies' health, random samples from different time, places and different groups of people will be collected to see how the overall health of the babies is like.



Motivation of sampling: the poll example

	DATES	POLLSTER	SAMPLE	RESULT	NET RESULT
U.S. Senate Miss.	NOV 25, 2018	C+ Change Research	1,211 LV	Espy 46% 51% Hyde-Smith	Hyde-Smith +5

Source: FiveThirtyEight.com

✱ This senate election poll tells us:

✱ The sample has 1211 likely voters

$$N = 1211$$

✱ Ms. Hyde-Smith has realized sample mean equal to 51%

✱ What is the estimate of the percentage of votes for Hyde-smith?

✱ How confident is that estimate?

Population

- ✱ What is a population?
 - ✱ It's the entire possible data set $\{X\}$
 - ✱ It has a countable size N_p
 - ✱ The population mean $popmean(\{X\})$ is a number
 - ✱ The population standard deviation is $popstd(\{X\})$ and is also a number
- ✱ The population mean and standard deviation are the same as defined previously in chapter 1

Population

$$\boxed{\{X\}} = \{1, 2, 3, \dots, 12\} \quad N_p = 12$$

$$\text{popmean}(\{X\}) = ? \quad \frac{13}{2}$$

$$\text{popstd}(\{X\}) = ?$$

$$\sqrt{\frac{1}{N_p} \sum_{i=1}^{12} (x_i - 6.5)^2}$$

$$N_p = 12$$

Sample

- ✱ The sample is a random subset of the population and is denoted as $\{x\}$, where sampling is done with **replacement**
- ✱ The sample size N is assumed to be much less than population size N_p
- ✱ The **sample mean of a population** is $\bar{X}^{(N)}$ and is a **random variable**

Sample $\{x\}$ and Sample Mean $X^{(N)}$

$$\{X\} = \{1, 2, 3, \dots, 12\} \quad N < N_p$$

One potential sample $\rightarrow \{x\} = \{1, 1, 2, 3, 3\} \quad N = 5$

$X^{(N)}$ RV takes a value?

$$X^{(N)} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

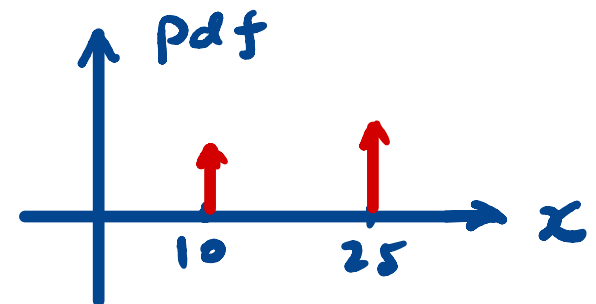
Another potential sample $\rightarrow \{1, 1, 1, 1, 1\} \Rightarrow X^{(N)} = ?$

Sample mean example



Money box

10¢ 25¢
dime quarter



* shake and take
one and put
back.

$$E(\bar{X}) = E(X)$$

X_1 takes $x_1 = 10$ $E[X] = ?$

X_2 takes $x_2 = 10$

X_3 takes $x_3 = 25$

$\vdots$
 X_N takes $x_N = 10$

$$\boxed{\bar{X}} = \frac{\sum X_i}{N}$$

" $X^{(N)}$ "

Sample mean of a population

- ✱ The sample mean is the average of **IID** samples

$$\underline{X^{(N)}} = \frac{1}{N}(X_1 + X_2 + \dots + X_N) = \text{mean}(\{x\})$$

- ✱ By linearity of the expectation and the fact the sample items are identically drawn from the same population with replacement

$$E[X^{(N)}] = \frac{1}{N}(E[X^{(1)}] + E[X^{(1)}].. + E[X^{(1)}]) = E[X^{(1)}]$$

Expected value of one random sample is the population mean

- ✱ Since each sample is drawn uniformly from the population

$$\sum x \cdot p(x) = \sum x \cdot \frac{1}{N_p} = \frac{1}{N_p} \cdot \sum x$$

$$E[X^{(1)}] = \text{popmean}(\{X\})$$

therefore $E[X^{(N)}] = \text{popmean}(\{X\})$

const

- ✱ We say that $X^{(N)}$ is an unbiased estimator of the population mean.

$$E[R_E] = \text{Target} + \text{Val}$$



R_E is random variable

const

Standard deviation of the sample mean

- ✱ We can also rewrite another result from the lecture on the weak law of large numbers

$$\text{var}[X^{(N)}] = \frac{\text{popvar}(\{X\})}{N}$$

Handwritten note: $\text{std}[X^{(N)}] = \sqrt{\text{var}[X^{(N)}]}$

- ✱ The standard deviation of the sample mean

$$\text{std}[X^{(N)}] = \frac{\text{popstd}(\{X\})}{\sqrt{N}}$$

- ✱ But we need the population standard deviation in order to calculate the $\text{std}[X^{(N)}]$!

Adjusted estimate of population standard deviation & the Stderr

✱ We use the following to estimate $popsd(\{X\})$

$$v = stdunbiased(\{x\}) = \sqrt{\frac{1}{N-1} \sum_{x_i \in \text{sample}} (x_i - \text{mean}(\{x_i\}))^2}$$

$\therefore E[v^2] = \text{var}[\{X\}]$

✱ Then the **standard error** is an estimate of

$$std[X^{(N)}]$$

recall : $std[X^{(N)}] = \frac{popsd(\{X\})}{\sqrt{N}}$

approximated by

$$\frac{popsd(\{X\})}{\sqrt{N}} \stackrel{\cdot}{=} \frac{stdunbiased(\{x\})}{\sqrt{N}} = \boxed{stderr(\{x\})}$$

Standard error: election poll

	DATES	POLLSTER	SAMPLE	RESULT	NET RESULT
U.S. Senate	Miss. NOV 25, 2018	C+ Change Research	1,211 LV	Espy 46% 51% Hyde-Smith	Hyde-Smith +5

✱ What is the estimate of the percentage of votes for Hyde-smith? 51%

$X_i = \begin{cases} 1 & \text{for Smith} \\ 0 & \text{otherwise} \end{cases}$

Number of sampled voters who selected Ms. Smith is:

$$1211(0.51) \approx 618$$

$$\sum X_i = \text{total votes for Smith}$$

Number of sampled voters who didn't selected Ms. Smith was

$$1211(0.49) \approx 593$$

$$\bar{X}^{(N)} = \frac{\sum_{i=1}^N X_i}{N}$$

$$N = 1211$$

Standard error: election poll

✱ $stdunbiased(\{x\})$

$$= \sqrt{\frac{1}{1211 - 1} (618(1 - 0.51)^2 + 593(0 - 0.51)^2)} = 0.5001001$$

$\{ 0, 0, \dots, 1, \dots \}$
 $N = 1211$

✱ $stderr(\{x\})$

$$= \frac{0.5001001}{\sqrt{1211}} \simeq 0.0144$$

Interpreting the standard error

- ✱ **Sample mean** is a random variable and has its own probability distribution, `stderr` is an estimate of sample mean's standard deviation $X^{(N)} \rightarrow \text{Normal if } N \rightarrow \infty$
- ✱ When **N** is very large, according to the **Central Limit Theorem**, sample mean is approaching a normal distribution with μ σ

$$\mu = \text{popmean}(\{X\}) ; \sigma = \frac{\text{popstd}(\{X\})}{\sqrt{N}} \doteq \text{stderr}(\{x\})$$

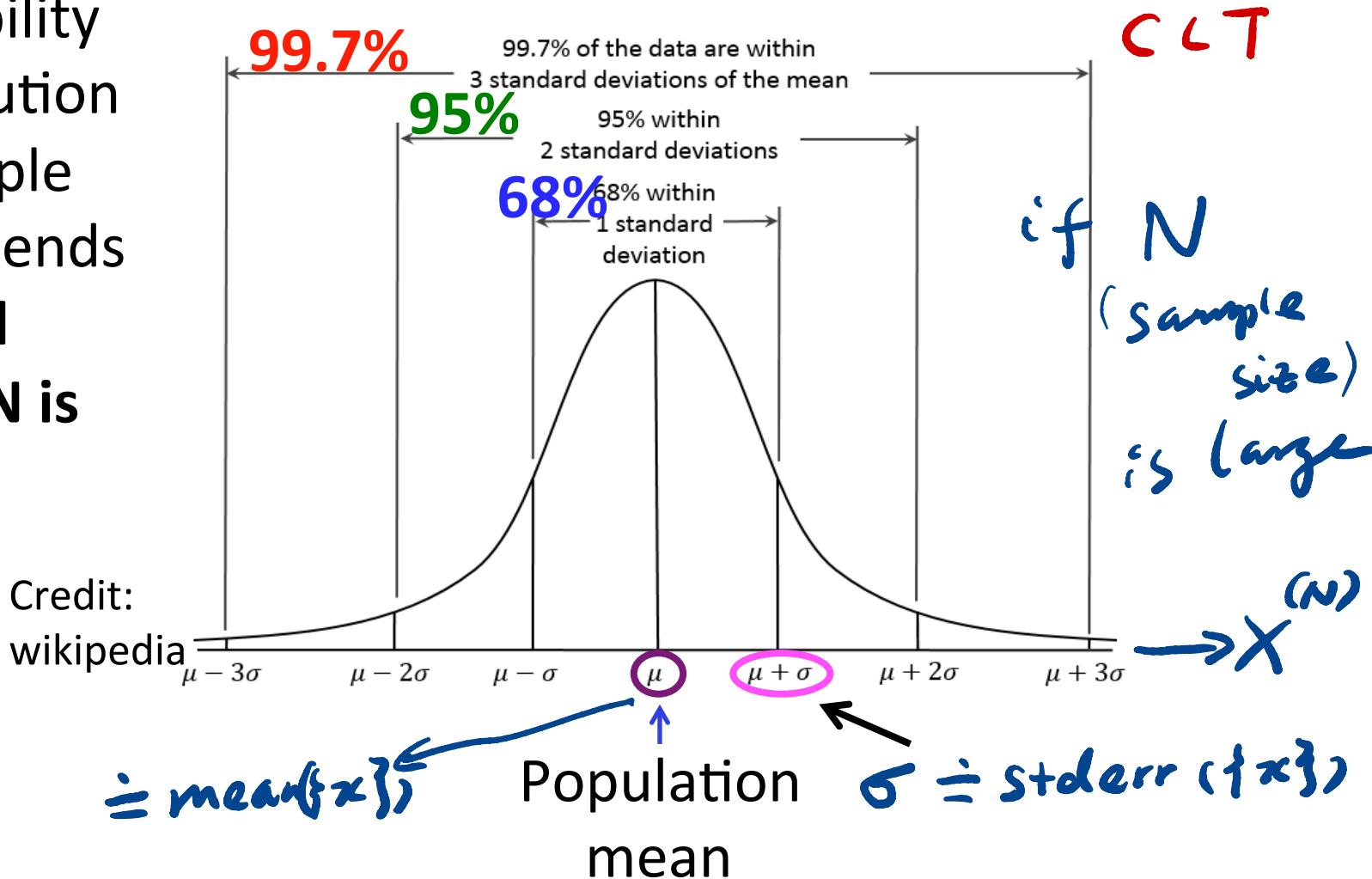
$$\bar{X} = \frac{\sum_{i=1}^N X_i}{N}$$

i.i.d.

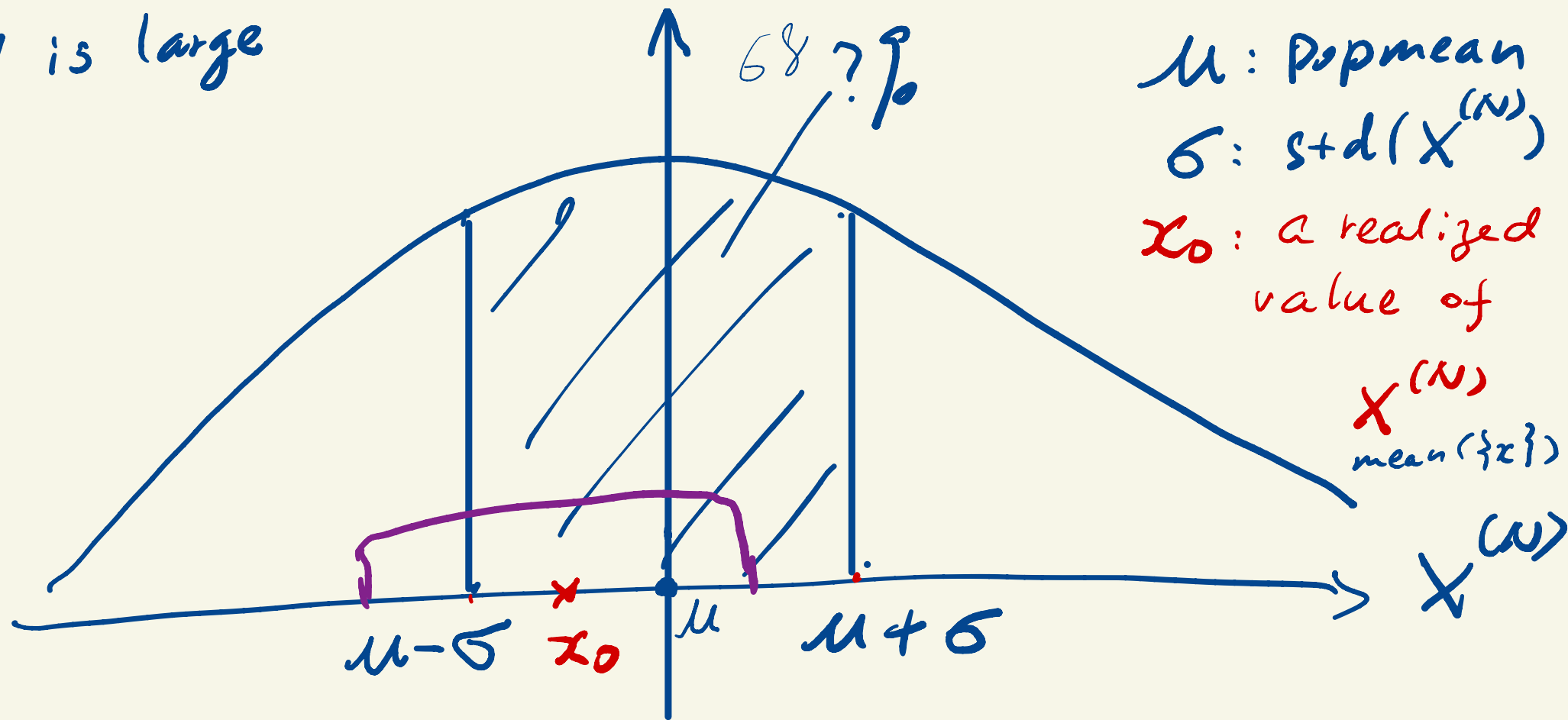
$$\text{stderr}(\{x\}) = \frac{\text{stdunbiased}(\{x\})}{\sqrt{N}}$$

Interpreting the standard error

Probability
distribution
of sample
mean tends
normal
when N is
large



N is large

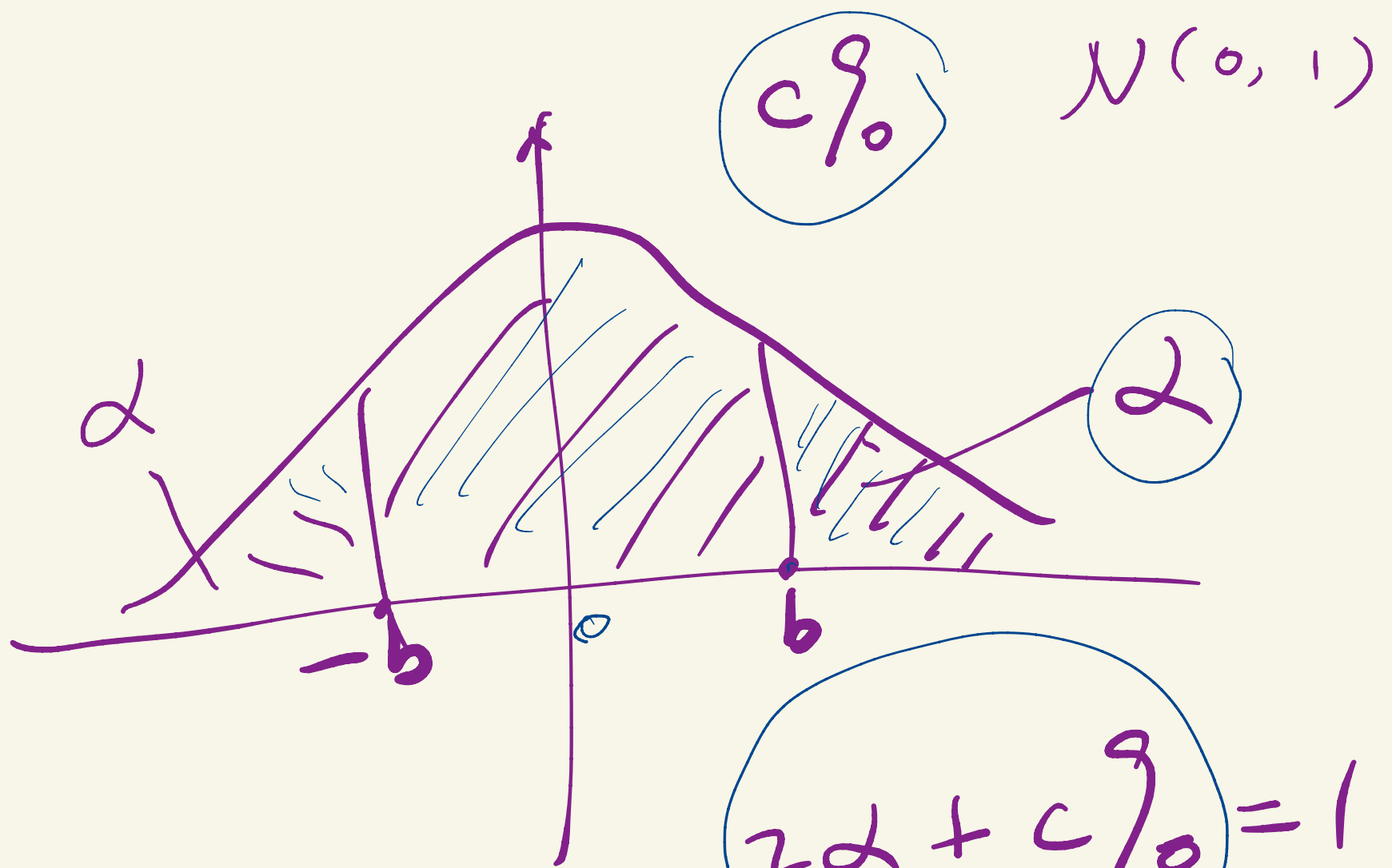


How much chance is it
that x_0 is

located in $[\mu - \sigma, \mu + \sigma]$?

If $\mu - \sigma < x_0 < \mu + \sigma$
then $\Rightarrow$

$$x_0 - \sigma < \mu < x_0 + \sigma$$



$X_0 \pm b \cdot \text{stderr}$

$$\underline{CDF(b) = 1 - \alpha}$$

$$X^{(N)} = \bar{X}$$

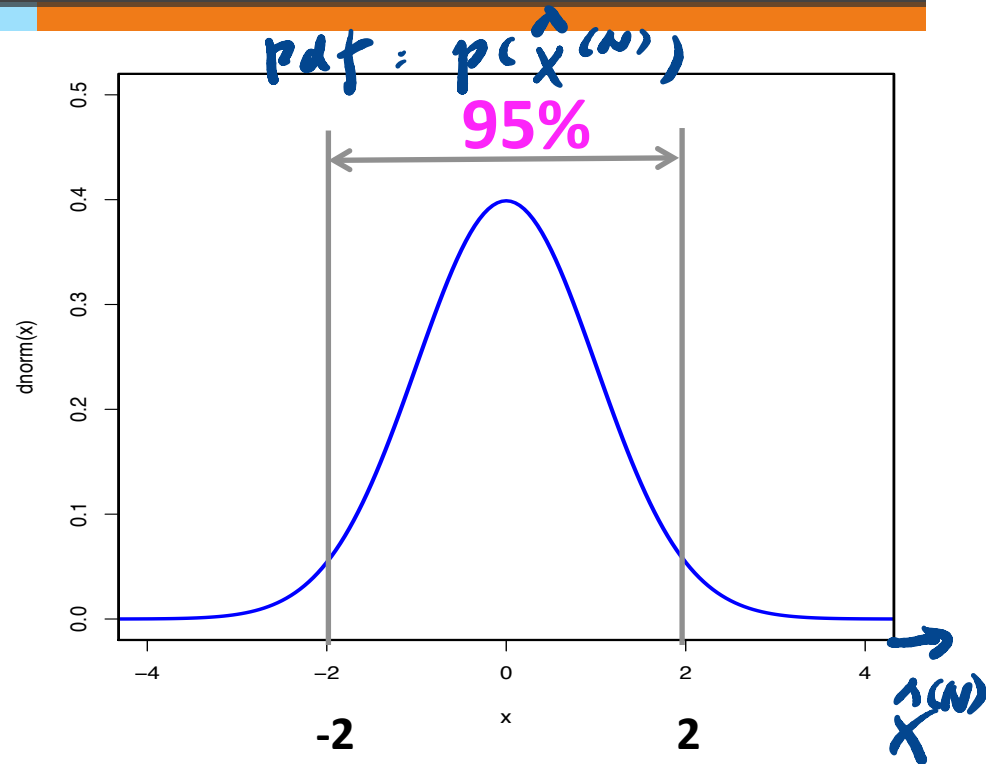
$$E[\bar{X}] = \text{pop mean}$$

$$E[\hat{\bar{X}}] = E\left(\frac{\bar{X} - \text{pop mean}}{\sigma(\bar{X})}\right) = 0$$

$$\sigma[\hat{\bar{X}}] = \sigma\left(\frac{\bar{X} - \text{pop mean}}{\sigma(\bar{X})}\right) = 1$$

Confidence intervals

- ✱ Confidence interval for a population mean is defined by fraction
- ✱ Given a percentage, find how many units of strerr it covers.



For **95%** of the **realized sample means**,
the population mean lies in
[sample mean - 2 stderr, sample mean + 2 stderr]

Confidence intervals when N is large

- ✱ For about 68% of realized sample means

$$\text{mean}(\{x\}) - \text{stderr}(\{x\}) \leq \text{popmean}(\{X\}) \leq \text{mean}(\{x\}) + \text{stderr}(\{x\})$$

- ✱ For about 95% of realized sample means

$$\text{mean}(\{x\}) - 2\text{stderr}(\{x\}) \leq \text{popmean}(\{X\}) \leq \text{mean}(\{x\}) + 2\text{stderr}(\{x\})$$

- ✱ For about 99.7% of realized sample means

$$\text{mean}(\{x\}) - 3\text{stderr}(\{x\}) \leq \text{popmean}(\{X\}) \leq \text{mean}(\{x\}) + 3\text{stderr}(\{x\})$$

Q. Confidence intervals

- ✱ What is the 68% confidence interval for a population mean?
- A. [sample mean-2stderr, sample mean+2stderr]
- ☒ B. [sample mean-stderr, sample mean+stderr]
- C. [sample mean-std, sample mean+std]

Standard error: election poll

	DATES	POLLSTER	SAMPLE	RESULT	NET RESULT
U.S. Senate Miss.	NOV 25, 2018	C+ Change Research	1,211 LV	Espy 46% 51% Hyde-Smith	Hyde-Smith +5

51%

✱ We estimate the population mean as 51% with stderr 1.44%

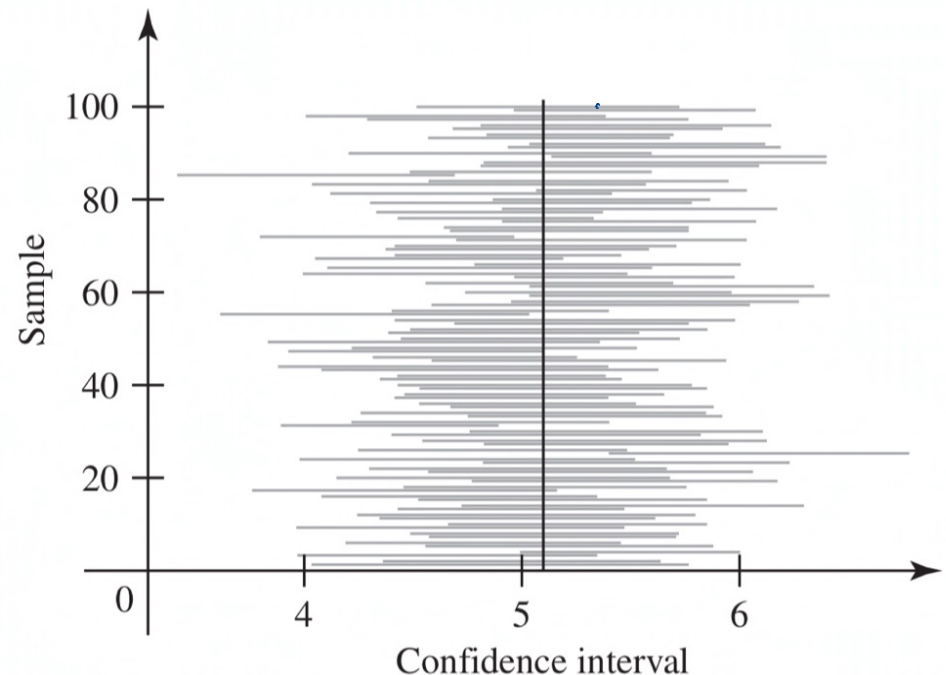
✱ The 95% confidence interval is
[51%-2×1.44%, 51%+2×1.44%] = [48.12%, 53.88%]

$$51\% \pm 2 \times \text{stderr}$$

Interpreting the confidence intervals

$$\text{mean}(\{x\}) \pm 2 \cdot \text{stdem}$$

Figure 8.5 A sample of one hundred observed 95% confidence intervals based on samples of size 26 from the normal distribution with mean $\mu = 5.1$ and standard deviation $\sigma = 1.6$. In this figure, 94% of the intervals contain the value of μ .



Deyoung Pg 487

Q.

✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. if I pick 30 apples and found 21 fuji , what is my 95% confidence interval to estimate the popmean is 70% for fuji? (hint: $\text{strerr} > 0.05$)

A. $[0.7-0.17, 0.7+0.17]$

B. $[0.7-0.056, 0.7+0.056]$

What if N is small? When is N large enough?

- ✱ If samples are taken from normal distributed population, the following variable is a random variable whose distribution is Student's t-distribution with **N-1** degree of freedom.

$$T = \frac{\text{mean}(\{x\}) - \text{popmean}(\{X\})}{\text{stderr}(\{x\})}$$

$E[\tau] = 0$
 $\text{var}[\tau] = 1$

$\nearrow x^{(N)}$
 $E[x^{(N)}]$

$x_1, \dots, x_N$

$\approx \rightarrow \text{std}(x^{(N)})$

N is the sample size

Degree of freedom is **N-1** due

to this constraint: $\sum_i (x_i - \text{mean}(\{x\})) = 0$

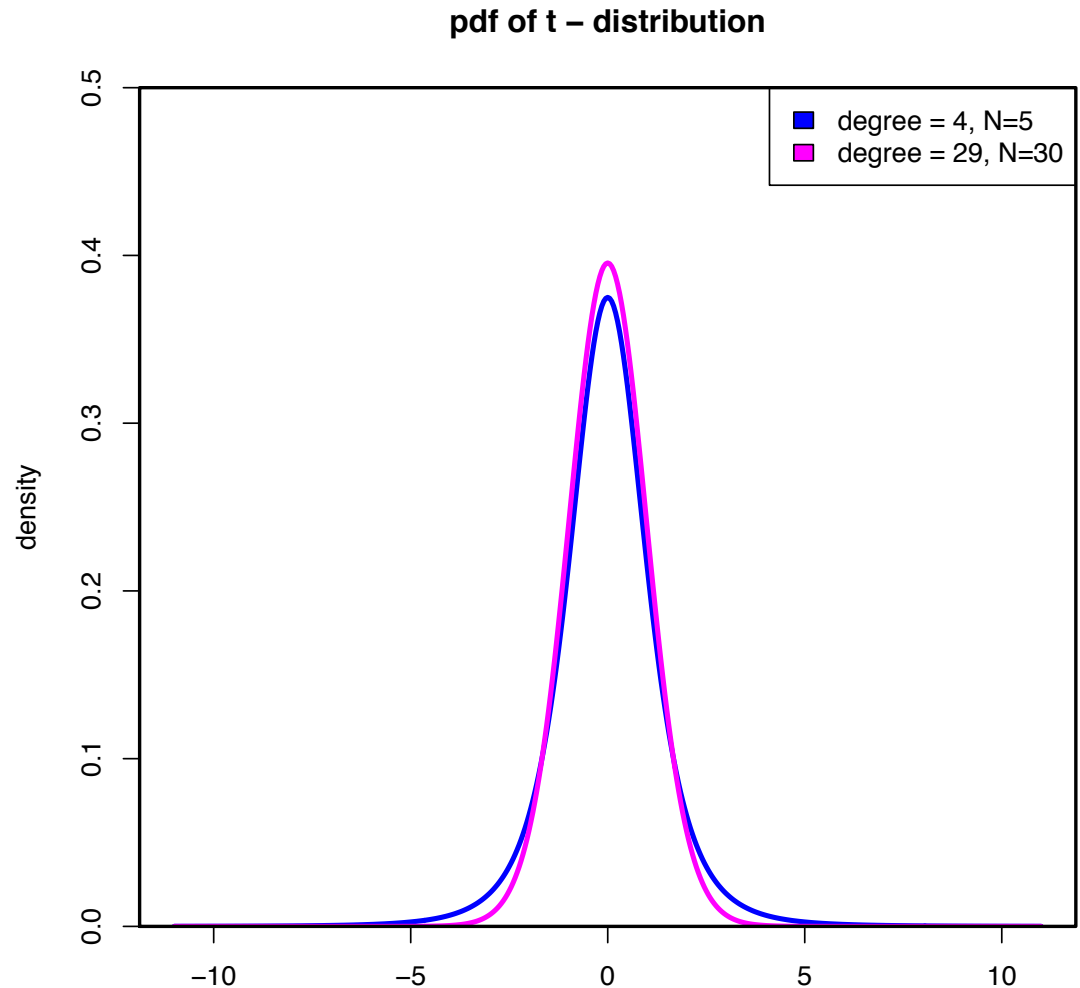
t-distribution is a family of distri. with different degrees of freedom

t-distribution with $N=5$
and $N=30$



Credit :
wikipedia

William Sealy Gosset 1876-1937

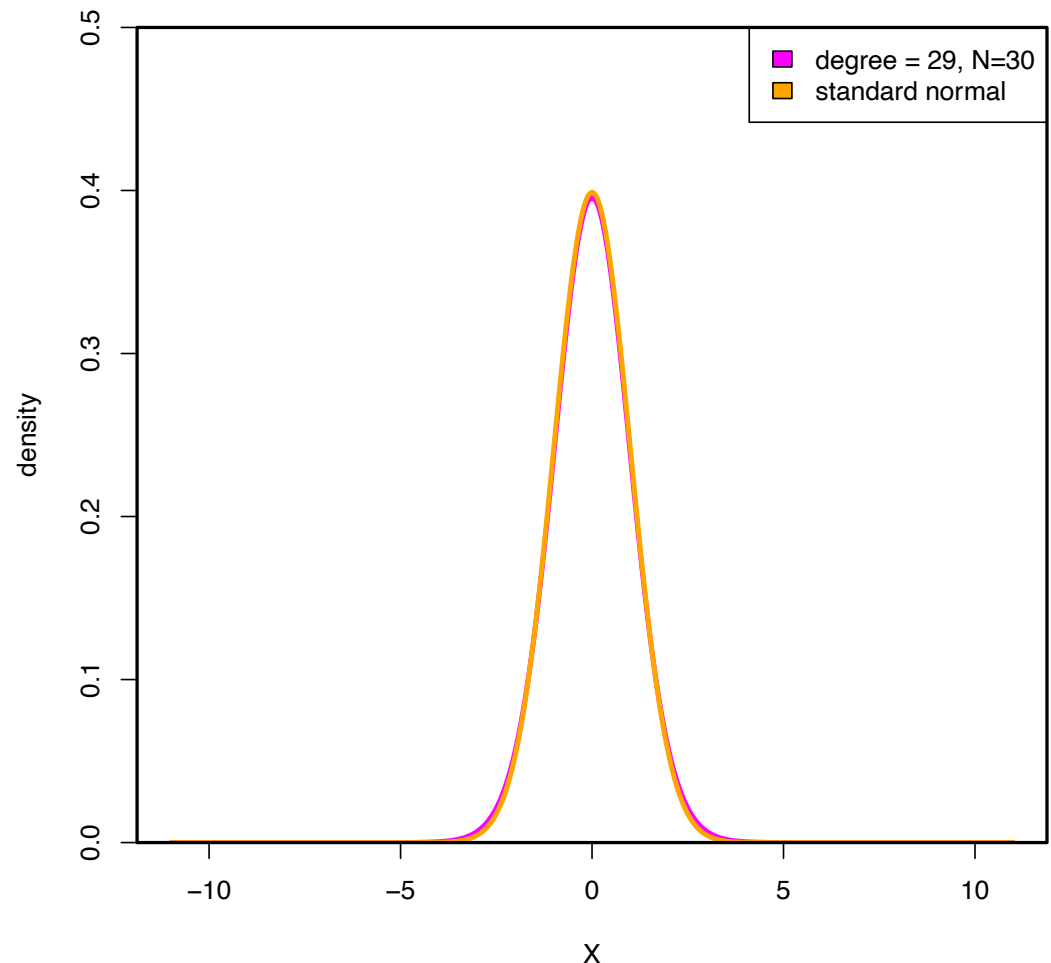


When $N=30$, t-distribution is almost Normal

t-distribution looks very similar to standard normal when $N=30$.

So, $N=30$ is a rule of thumb to decide N is large or not

pdf of t (n=30) and normal distribution



Confidence intervals when $N < 30$

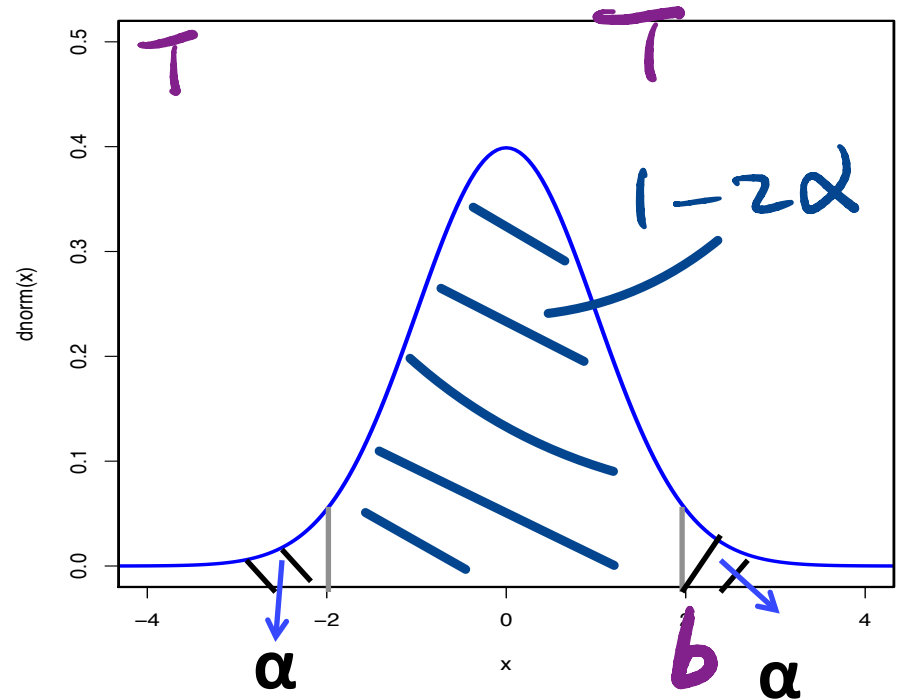
- ✱ If the sample size $N < 30$, we should use t-distribution with its parameter (**the degrees of freedom**) set to $N-1$

Centered Confidence intervals

- ✱ Centered Confidence interval for a population mean by α value, where

$$P(T \geq b) = \alpha$$

$$P(|T| \geq b) = 2\alpha$$



For $1-2\alpha$ of the realized sample means,
the population mean lies in
[sample mean- $b \times \text{stderr}$, sample mean+ $b \times \text{stderr}$]

2α Confidence Interval

$$P\left(\frac{\text{mean}(\{x\}) - \text{popmean}}{\text{stderr}(\{x\})} \geq b\right) = \alpha$$

$$= P(\text{popmean} \leq \text{mean}(\{x\}) - b \cdot \text{stderr}(\{x\})) = \alpha$$

$$P\left(\frac{\text{mean}(\{x\}) - \text{popmean}}{\text{stderr}(\{x\})} \leq -b\right) = \alpha$$

$$= P(\text{popmean} \geq \text{mean}(\{x\}) + b \cdot \text{stderr}(\{x\}))$$

$$= \alpha$$

$$\alpha = 5\% \rightarrow 1 - 2\alpha = 90\%$$

Q.

✱ The 95% confidence interval for a population mean is equivalent to what $1-2\alpha$ interval?

A. $\alpha = 0.05$

B. $\alpha = 0.025$

C. $\alpha = 0.1$

Assignments

- ✱ Read Chapter 7 of the textbook
- ✱ Next time: Bootstrap, Hypothesis tests
- ✱ Prepare for Midterm1

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

See you next time

*See
you!*

